

Supplementary Material

Anonymous WACV [Datasets Track](#) submission

Paper ID 977

A. Supplementary Materials

This supplement expands the benchmark construction, perturbation characterization, matched marginal-joint analyses, statistical sensitivity tests, recovery experiments, and release protocol supporting TandemBench. The emphasis follows the revised main paper: all twelve text perturbations belong to the benchmark; the eight operators with mean fire rate above 55% form the execution-screened primary analysis subset; and a nested seven-operator high-damage subset is used only as a challenge/stress-test condition. Marginal damage is treated as benchmark difficulty rather than as a validity criterion.

The supplement also reports analyses that are intentionally too detailed for the main paper: positioning against prior robustness evaluations, complete checkpoint and severity specifications, operator-set sensitivity, held-out screening checks, symmetric image-side sensitivity, damage-matched analysis, metric-resolved additive composition, crossed-axis bootstrap sensitivity, complete recovery experiments, and an expanded version of the released evaluation protocol.

B. Positioning Against Prior Robustness Evaluations

Prior work has already studied multimodal corruption, including settings in which both modalities are degraded simultaneously. The distinction made by TandemBench is therefore not “joint corruption exists.” It is the matched measurement design: every joint retrieval condition is paired with its corresponding image-only and text-only conditions on the same samples and clean baseline, producing $(\delta_1, \delta_2, \delta_3)$ for explicit composition analysis. Table S1 makes that narrower distinction visible.

C. Benchmark Assets and Evaluation Setup

C.1. Datasets and retrieval protocol

TandemBench uses the 5,000-image MS-COCO val2017 split with 25,014 captions and the 1,000-image Flickr30K

Karpathy test split with 5,000 captions. No gallery subsampling is used. Both image-to-text (I2T) and text-to-image (T2I) retrieval are evaluated. Recall@1 is the primary metric; Recall@5 and Recall@10 are additionally used to test whether the composition findings depend on retrieval cut-off.

Every corruption effect is a drop from the clean score for the same model, dataset, direction, and metric. Path 1 defines image-side degradation δ_1 , Path 2 defines text-side degradation δ_2 , and Path 3 defines joint degradation δ_3 .

COCO retrieval is often reported on the Karpathy 5K test split rather than val2017. Absolute clean scores should therefore be interpreted on the split used here. Corruption effects remain internally matched because every degradation is computed against the clean baseline on that same split.

C.2. Models, checkpoints, and clean baselines

The roster contains 19 dual-encoder retrieval models from eight families. BLIP-large is evaluated without image-text-matching reranking so that all models use independent image/text embeddings and the same similarity-based retrieval protocol. BLIP-large and CoCa ViT-L are COCO-finetuned checkpoints. OpenCLIP ViT-B/16 uses LAION-400M, whereas the remaining OpenCLIP variants use LAION-2B tags; CLIP, MetaCLIP, EVA-CLIP, and SigLIP keep their pretraining data source fixed within the variants evaluated here.

C.3. Evaluation accounting and analysis subsets

The release contains 83,754 non-null evaluation records: 6,156 image-only records, 4,560 text-only records, and 73,038 jointly corrupted records. The basic analysis unit is a complete matched triple $(\delta_1, \delta_2, \delta_3)$ indexed by model, dataset, retrieval direction, image operator, text operator, and severity.

Image and text operators are paired by a common ordinal severity index $s \in \{1, \dots, 5\}$, tracing the diagonal of the full image-text severity grid. Equal indices order intervention strength within each operator; they do not imply equal physical, linguistic, perceptual, or realized retrieval damage

Table S1. Positioning against prior robustness evaluations. “Joint” denotes simultaneous degradation of both modalities of the same example; “Matched marginals” denotes corresponding image-only and text-only measurements; “Composition” denotes explicit analysis of how the joint effect relates to those two marginals.

Work	Joint	Natural corr.	Matched marginals	Composition	Retrieval
ImageNet-C [6]	✗	✓	✗	✗	✗
MM-Robustness [13]	✗	✓	✗	✗	✓
Composed-retrieval [18]	✗	✓	✗	✗	✓
Adaptation robustness [3]	✗	✓	✗	✗	✗
MMCBench [22]	✗	✓	✗	✗	✗
Video-text [16]	✗	✓	✗	✗	✗
R-Bench [9]	✗	✓	✗	✗	✗
VLM-RobustBench [15]	✗	✓	✗	✗	✗
VLSRB [8]	✓	✓	✗	✗	✗
Mint [1]	✗	✓	✗	✗	✗
TCR [10]	✗	✓	✗	✗	✓
UCA [24]	✗	✓	✗	✗	✓
MG-MTTA [2]	✓	✓	✗	✗	✗
Adversarial [11, 21]	✓	✗	✗	✗	✓
TandemBench (ours)	✓	✓	✓	✓	✓

Table S2. Source datasets and retrieval setup.

Dataset	Images	Captions	Directions
MS-COCO val2017	5,000	25,014	I2T/T2I
Flickr30K Karpathy test	1,000	5,000	I2T/T2I

across operators. The main paper therefore treats realized-damage matching as a separate post-hoc sensitivity analysis rather than as a physical calibration of the severity scale.

D. Corruption Suite and Severity Definitions

D.1. Image corruptions

We evaluate 16 image corruptions in four families: noise (Gaussian, shot, impulse, speckle), blur (defocus, glass, motion, zoom), weather (snow, frost, fog, brightness), and digital corruption (contrast, elastic transform, pixelate, JPEG compression). Fifteen are standard ImageNet-C corruptions [6]; speckle noise comes from the extended corruption set. All image operators are defined for every image and therefore require no structural applicability check.

Severity is ordinal within an operator rather than a shared unit across operators. The same generated corrupted images are reused across models for a given dataset, operator, and severity.

D.2. Text perturbations

The 12 text perturbations cover character-level, word-level, and semantic/structure-dependent edits following common NLP robustness taxonomies and toolkits [12, 14, 20].

Requested rate and realized execution are intentionally separated. A higher requested severity cannot create linguistic structure that is absent from a short caption. For `sentence_shuffle`, benchmark applicability is defined by either two or more sentences or multiple comma/semicolon-separated phrases; this yields 5.2% ap-

plicability in the benchmark characterization.

D.3. Qualitative full-suite gallery

E. Perturbation Execution, Applicability, and Difficulty

The revised benchmark separates three quantities that answer different questions: *applicability* asks whether the structure required by an operator exists, *fire rate* asks whether the requested transformation actually changes the input, and *marginal difficulty* measures the resulting retrieval degradation. Damage is not used to decide whether an operator belongs to TandemBench.

E.1. Complete operator characterization

For every text operator and severity, 500 captions are perturbed and fire rate is the fraction whose output string differs from the input. Structure-dependent operators also receive an applicability estimate. The eight operators with severity-averaged fire rate above 55% form the primary execution-screened subset. A nested seven-operator challenge subset additionally requires more than 8 Recall@1 points of text-side marginal damage.

`punctuation_noise` is the clearest separation between execution and difficulty: it fires on 56.0% of captions on average and 90% at severity 5, yet induces only 1.25 points of text-side degradation. It therefore belongs to the primary execution subset even though it is not in the high-damage challenge subset. Conversely, the four low-fire semantic operators are constrained by the structures present in short captions. All twelve remain part of the benchmark suite.

E.2. Rate-limited versus structure-limited execution

Broadly applicable character/word operators increase sharply with requested severity; for example, `char_swap`

Table S3. Complete model roster, checkpoint tags, parameter counts, and clean Recall@1 baselines. A dash indicates a loader-specific checkpoint rather than an `open.clip` pretrained tag. BLIP is evaluated without image–text-matching reranking.

Family	Model key	M	Pretrained tag	COCO I2T	COCO T2I	F30K I2T	F30K T2I
CLIP	clip_vit_b32	151	–	49.96	30.38	78.80	58.78
CLIP	clip_vit_b16	150	–	51.72	32.70	82.20	62.14
CLIP	clip_vit_l14	428	–	56.00	35.33	85.20	64.86
CLIP	clip_vit_l14_336	428	–	57.54	36.10	87.70	66.90
OpenCLIP	openclip_vit_b32	151	laion2b_s34b_b79k	56.56	38.79	84.20	66.76
OpenCLIP	openclip_vit_b16	150	laion400m_e32	56.48	37.90	83.50	65.64
OpenCLIP	openclip_vit_l14	428	laion2b_s32b_b82k	62.66	45.70	89.50	75.52
OpenCLIP	openclip_vit_h14	986	laion2b_s32b_b79k	66.16	48.50	90.70	77.62
OpenCLIP	openclip_vit_big14	2540	laion2b_s39b_b160k	66.94	50.68	92.90	79.58
MetaCLIP	metaclip_vit_b16	150	metaclip_fullcc	58.10	41.36	85.50	70.72
MetaCLIP	metaclip_vit_l14	428	metaclip_fullcc	64.52	46.63	90.00	76.44
EVA-CLIP	evaclip_vit_b16	150	merged2b_s8b_b131k	58.74	41.66	86.10	71.38
EVA-CLIP	evaclip_vit_l14	428	merged2b_s4b_b131k	63.82	46.86	89.70	77.28
SigLIP	siglip_vit_b16	203	–	65.30	47.19	89.10	74.60
SigLIP	siglip_vit_l16	652	–	70.38	51.48	91.70	79.12
SigLIP	siglip_so400m	878	–	72.08	53.45	94.20	83.06
CoCa	coca_vit_l	638	mscoco_finetuned_laion2B-s13B-b90k	67.58	53.07	88.90	78.70
DFN	dfn_vit_h14	986	dfn5b	71.10	53.24	92.90	80.14
BLIP	blip_large	470	–	79.64	62.73	93.60	82.10

Table S4. Matched-condition accounting under the revised analysis policy. Non-normalized analyses use all complete triples. The 0.5-pp safeguard is applied only to normalized ratio analyses.

Text-operator set	Operators	Nominal cells	Complete triples	After $\delta_1 + \delta_2 \geq 0.5$ pp
Full benchmark suite	12	72,960	71,040	69,435
Execution-screened primary	8	48,640	47,360	46,949
High-damage challenge	7	42,560	41,440	41,264

Table S5. Exact severity schedule for the 16 image corruptions. Parameters follow the defaults used by the released `imagecorruptions` generator. For glass blur, (σ, d, i) denotes blur width, local displacement, and iterations; for elastic transform, $\sigma = 0.01$ of image shape and maximum displacement is 0.005 of image height. Zoom entries denote start:step:end. The snow tuple is the implementation’s seven-parameter snow-layer configuration.

Operator	Parameter	S1	S2	S3	S4	S5
Gaussian noise	σ	.08	.12	.18	.26	.38
Shot noise	λ	60	25	12	5	3
Impulse noise	p	.03	.06	.09	.17	.27
Speckle noise	σ	.15	.20	.35	.45	.60
Defocus blur	(r, a)	(3.,1)	(4.,5)	(6.,5)	(8.,5)	(10.,5)
Glass blur	(σ, d, i)	(7,1,2)	(9,2,1)	(1,2,3)	(1,1,3,2)	(1,5,4,2)
Motion blur	(r, σ)	(10,3)	(15,5)	(15,8)	(15,12)	(20,15)
Zoom blur	zoom factors	1:01:1.10	1:01:1.15	1:02:1.20	1:02:1.25	1:03:1.30
Snow	7-tuple	(.1, .3, 3, .5, 10, 4, .8)	(.2, .3, 2, .5, 12, 4, .7)	(.55, .3, 4, .9, 12, 8, .7)	(.55, .3, 4, 5, .85, 12, 8, .65)	(.55, .3, 2, 5, .85, 12, 12, .55)
Frost	blend (α, β)	(1,4)	(8.,6)	(7.,7)	(65.,7)	(6.,75)
Fog	(s, w)	(1.5,2)	(2,2)	(2.5,1.7)	(2.5,1.5)	(3,1.4)
Brightness	HSV offset	.1	.2	.3	.4	.5
Contrast	factor	.4	.3	.2	.1	.05
Elastic transform	α (250 px ref.)	12.5	16.25	21.25	25	30
Pixelate	rescale	.60	.50	.40	.30	.25
JPEG compression	quality	25	18	15	10	7

risers from 28% firing at severity 1 to 98% at severity 5. Structure-dependent operators remain bounded by caption content: `relation_swap` rises only from 11% to 41% against a 53.4% applicability ceiling, and `sentence_shuffle` does not fire in the sampled characterization despite 5.2% benchmark-level structural applicability.

E.3. Caption-length control and the sentence-shuffle eligibility correction

A caption-length control concatenates five human captions for the same COCO image while keeping the image and operator implementation fixed. For the semantic operators whose eligibility definitions are unchanged by the control, applicability increases sharply:

This control supports a narrow conclusion: short-caption structure constrains whether some semantic transformations

Table S6. Text-perturbation definitions and exact severity schedules. For semantic operators, the rate is applied only to eligible structures; requested severity therefore does not guarantee that a caption can be modified.

Operator	Class	Requires	Rate unit	S1	S2	S3	S4	S5
word_shuffle	surface	2+ tokens	tokens	10%	20%	35%	50%	60%
word_delete	surface	2+ tokens	tokens	5%	10%	20%	30%	40%
char_delete	surface	characters	characters	1%	3%	5%	7%	10%
char_swap	surface	characters	characters	1%	3%	5%	7%	10%
object_substitution	semantic	a noun	eligible tokens	10%	25%	40%	60%	80%
keyboard_typo	surface	characters	characters	1%	3%	5%	7%	10%
ocr_error	surface	characters	characters	1%	3%	5%	7%	10%
punctuation_noise	surface	terminal punctuation	punctuation	5%	10%	20%	30%	40%
relation_swap	semantic	a spatial preposition	eligible tokens	10%	25%	40%	60%	80%
attribute_swap	semantic	an adjective	eligible tokens	10%	25%	40%	60%	80%
negation_insertion	semantic	a verb or copula	eligible tokens	10%	25%	40%	60%	80%
sentence_shuffle	semantic	2+ sentences or comma/semicolon phrases	sentences	20%	35%	50%	65%	80%

Table S7. Complete text-perturbation characterization. Exec. denotes membership in the execution-screened primary subset; Chal. denotes the nested high-damage challenge subset. Marginal damage is difficulty, not a validity criterion.

Operator	Class	Fire %	S5 %	Eligible %	δ_2 pp	Exec.	Chal.
word_shuffle	surface	99.6	100	100.0*	8.82	✓	✓
word_delete	surface	75.0	99	100.0*	11.69	✓	✓
char_delete	surface	74.4	98	100.0*	11.68	✓	✓
char_swap	surface	74.0	98	100.0*	14.55	✓	✓
object_substitution	semantic	70.6	98	100.0	12.57	✓	✓
keyboard_typo	surface	69.8	97	100.0*	12.01	✓	✓
ocr_error	surface	59.8	90	100.0*	8.67	✓	✓
punctuation_noise	surface	56.0	90	100.0*	1.25	✓	–
relation_swap	semantic	25.6	41	53.4	0.32	–	–
attribute_swap	semantic	18.2	30	50.9	3.89	–	–
negation_insertion	semantic	4.0	9	41.3	0.65	–	–
sentence_shuffle	semantic	0.0	0	5.2	0.26	–	–

* Applicability is 100% by construction.

Table S8. Caption-length applicability control. The sentence_shuffle row is intentionally omitted because the historical control used a stricter “two sentences only” definition; the benchmark uses the broader implemented criterion described above.

Operator	Eligible (%)		Fires (%)	
	Single	Concatenated	Single	Concatenated
negation_insertion	35.9	93.4	18.25	69.75
attribute_swap	52.3	93.9	27.50	73.75
relation_swap	54.5	90.7	24.75	67.00
object_substitution	100.0	100.0	79.50	100.00

can act. It is not a human semantic-naturalness audit. For sentence_shuffle, the benchmark-level 5.2% applicability value supersedes the earlier 0.2% sentence-only control and is the value used in the main paper.

E.4. Inspectable perturbation outputs

The release includes original and corrupted captions for the screening runs, allowing fire rate to be audited at the individual-caption level rather than inferred from retrieval performance.

F. Operator-Set and Screening Sensitivity

F.1. Full suite, execution-screened primary subset, and challenge subset

The central result does not require outcome-based operator selection. Table S9 reports all complete, unfiltered triples for the non-normalized quantities used in the headline analysis.

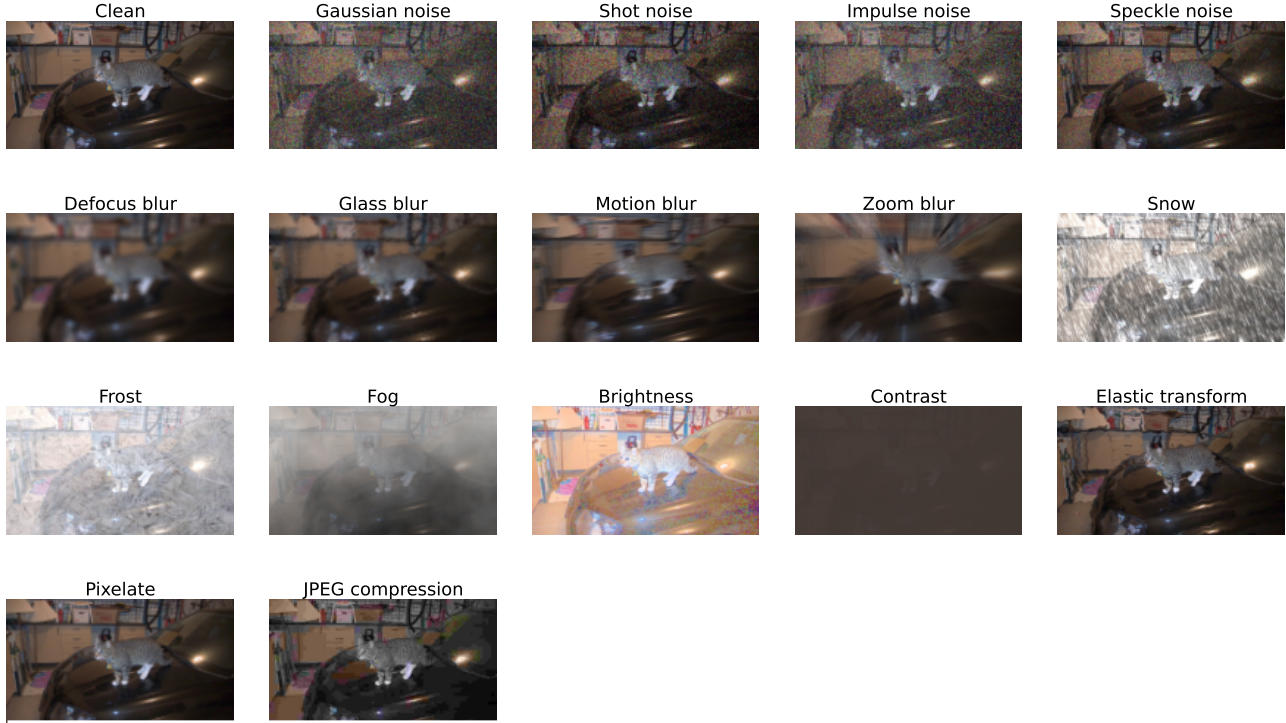
Selecting stronger operators increases the magnitude of the effect, but does not create it. The full twelve-operator suite already shows joint degradation above the larger marginal in 91.84% of complete conditions. The nested challenge subset is therefore reported as a stress test rather than as a definition of perturbation validity.

F.2. Held-out stability of the high-damage challenge screen

The > 8 -point damage criterion is outcome-dependent, so we test whether it selects a fragile operator set. Selecting the challenge operators on COCO and evaluating on Flickr, selecting on Flickr and evaluating on COCO, and selecting on either half of the model roster all recover the same seven operators. Held-out results remain large:

This analysis does not turn marginal damage into a validity criterion; it only shows that the secondary challenge

(a) Image corruptions: common source image, severity 5



(b) Text perturbations: severity-5 outputs

Clean caption: "A dog is running across the green field."

Keyboard Typo

Q dog is running zctoss ths green field.

Ocr Error

A dog is runnIng @cross the green field.

Char Swap

Aodg i srunninga cors stheg rene field.

Char Delete

dogis runnn cros he green field.

Word Delete

is across green

Word Shuffle

is across A field. dog green the running

Punctuation Noise

)A dog- ;is running across the green field.

Negation Insertion

A dog is not running never across the green field.

Attribute Swap

A dog is running across the blue field.

Relation Swap

A dog is running above the green field.

Object Substitution

A frump is running across the green area.

Sentence Shuffle (unchanged)

A dog is running across the green field.

Figure S1. Qualitative overview of the complete TandemBench corruption suite at severity 5. (a) The same source image is shown clean and under each of the 16 image corruption operators. (b) The same clean caption is passed through all 12 text operators at severity 5. `sentence_shuffle` remains unchanged because the one-sentence caption is structurally ineligible; this is part of the screening phenomenon rather than a missing visualization.

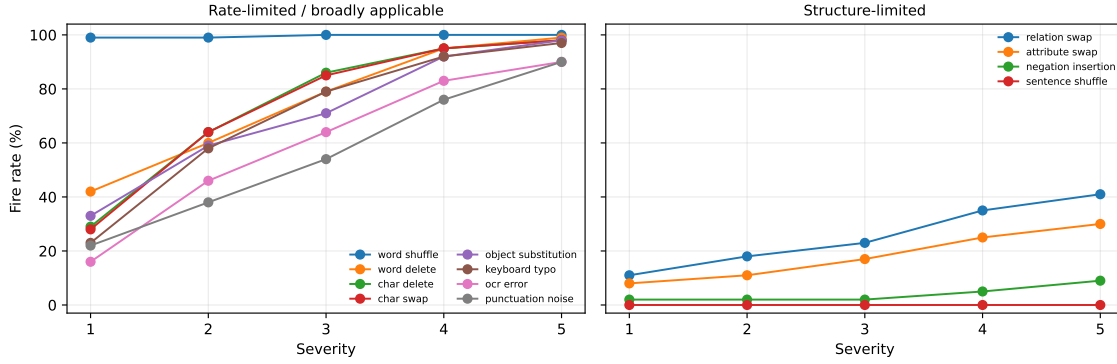


Figure S2. Realized text-perturbation fire rate across severity. Broadly applicable operators (left) generally approach saturation as the requested rate increases, whereas structure-dependent operators (right) remain bounded by the linguistic structures present in short retrieval captions.

Table S9. Sensitivity to text-operator composition. Additive error is $\delta_3 - (\delta_1 + \delta_2)$. No 0.5-pp filter is applied to these non-normalized quantities.

Subset	Text ops	Complete n	Exceed %	Mean excess	Mean additive error
Full suite	12	71,040	91.84	+3.79	-0.62
Execution-screened primary	8	47,360	95.15	+5.07	-0.99
High-damage challenge	7	41,440	96.15	+5.63	-1.13

Table S10. Held-out sensitivity of the seven-operator high-damage challenge screen. The challenge subset is secondary; this table only tests whether its composition is unstable to the data used for selection.

Selection $\rightarrow$ evaluation	Exceed %	Mean excess
COCO $\rightarrow$ Flickr30K	95.29	+6.52
Flickr30K $\rightarrow$ COCO	96.95	+4.77
Model half A $\rightarrow$ half B	96.02	+5.84
Model half B $\rightarrow$ half A	96.26	+5.42

subset is not selected by one unusually favorable split.

F.3. Symmetric image-side damage-screen sensitivity

As an additional sensitivity check, applying the same > 8 -point marginal-damage threshold on the image side retains 10 of 16 image operators. Under that symmetric restriction, the execution-screened text subset yields 97.46% exceedance and +6.23 Recall@1 points of mean excess; the seven-operator challenge subset yields 98.60% and +6.95 points. The main benchmark does not use this symmetric restriction because image operators are universally executable and marginal damage is treated as difficulty, but the result shows that the qualitative conclusion is not created by an asymmetric high-damage text screen.

F.4. Broader-pool execution audit

The execution/applicability logic is also applied to a broader pool of perturbations from published retrieval-robustness

suites. This is not an external gold-standard validation; it is an audit of whether the execution screen selectively rejects standard published operators while preserving only our preferred transformations.

Executed failures are concentrated in the structure-dependent semantic transformations, while standard structure-independent perturbations generally execute broadly. Generation-based whole-sentence rewrites that require additional models are listed as not run rather than counted as passes or failures.

G. Extended Joint-Corruption Results

G.1. Severity-resolved primary results

Table S12 uses the eight-operator execution-screened primary subset and all complete triples. Exceedance and excess increase with severity, while the fraction of conditions above the additive sum decreases.

The larger-marginal statistic is therefore best interpreted as degradation missed by single-modality evaluation, not as a claim of universal synergy. Joint degradation exceeds $\delta_1 + \delta_2$ in only 39.29% of primary conditions overall.

G.2. Dataset and retrieval direction

All four strata retain the same qualitative conclusion. Exceedance frequency and excess magnitude are distinct summaries: Flickr30K I2T has the lowest exceedance frequency but the largest mean excess.

Table S11. Complete 41-operator broader-pool screen. Fire rate and character-error rate (CER) are shown for executed operators; nine generation-based or trivially applying variants were not run. The four failures are TandemBench’s own structure-dependent operators.

Suite	Operator	Class	Run	Fire %	CER	Outcome
MM-Robustness	KeyboardAug	structure-independent	yes	100.0	0.1719	pass
MM-Robustness	OcrAug	structure-independent	yes	82.0	0.0899	pass
MM-Robustness	RandomCharAug_delete	structure-independent	yes	100.0	0.1722	pass
MM-Robustness	RandomCharAug_insert	structure-independent	yes	100.0	0.1724	pass
MM-Robustness	RandomCharAug_substitute	structure-independent	yes	100.0	0.1695	pass
MM-Robustness	RandomCharAug_swap	structure-independent	yes	98.2	0.1993	pass
MM-Robustness	EDA_synonym_replace	structure-independent	yes	100.0	0.3511	pass
MM-Robustness	EDA_random_insert	structure-independent	yes	99.8	0.4425	pass
MM-Robustness	EDA_random_swap	structure-independent	yes	98.8	0.3681	pass
MM-Robustness	EDA_random_delete	structure-independent	yes	84.6	0.3632	pass
TCR	KeyboardAug	structure-independent	yes	100.0	0.1719	pass
TCR	OcrAug	structure-independent	yes	82.0	0.0899	pass
TCR	RandomCharAug_delete	structure-independent	yes	100.0	0.1722	pass
TCR	RandomCharAug_insert	structure-independent	yes	100.0	0.1724	pass
TCR	RandomCharAug_substitute	structure-independent	yes	100.0	0.1695	pass
TCR	RandomCharAug_swap	structure-independent	yes	98.2	0.1993	pass
TCR	EDA_synonym_replace	structure-independent	yes	100.0	0.3511	pass
TCR	EDA_random_insert	structure-independent	yes	99.8	0.4425	pass
TCR	EDA_random_swap	structure-independent	yes	98.8	0.3681	pass
TCR	EDA_random_delete	structure-independent	yes	84.6	0.3632	pass
MM-Robustness	back_translation	structure-independent	no	–	–	not run
MM-Robustness	style_formal	structure-independent	no	–	–	not run
MM-Robustness	style_casual	structure-independent	no	–	–	not run
MM-Robustness	style_active	structure-independent	no	–	–	not run
MM-Robustness	style_passive	structure-independent	no	–	–	not run
MM-Robustness	insert_punctuation	structure-independent	no	–	–	not run
TCR	back_trans	structure-independent	no	–	–	not run
TCR	active/passive/casual/formal	structure-independent	no	–	–	not run
TCR	insert_punctuation	structure-independent	no	–	–	not run
ours	attribute_swap	structure-dependent	yes	18.2	0.0257	fail
ours	char_delete	structure-independent	yes	74.4	0.0508	pass
ours	char_swap	structure-independent	yes	74.0	0.0885	pass
ours	keyboard_typo	structure-independent	yes	69.8	0.0425	pass
ours	negation_insertion	structure-dependent	yes	4.0	0.0052	fail
ours	object_substitution	structure-independent	yes	70.6	0.2167	pass
ours	ocr_error	structure-independent	yes	59.8	0.0301	pass
ours	punctuation_noise	structure-independent	yes	56.0	0.0226	pass
ours	relation_swap	structure-dependent	yes	25.6	0.0374	fail
ours	sentence_shuffle	structure-dependent	yes	0.0	0.0000	fail
ours	word_delete	structure-independent	yes	75.0	0.2273	pass
ours	word_shuffle	structure-independent	yes	99.6	0.4718	pass

Table S12. Severity-resolved Recall@1 results on the execution-screened primary subset. Additive error is $\delta_3 - (\delta_1 + \delta_2)$.

S	n	Exceed %	Mean excess	$\bar{\delta}_1$	$\bar{\delta}_2$	$\bar{\delta}_3$	Joint > sum %
1	9,472	88.64	+1.34	2.79	2.17	5.11	55.00
2	9,472	92.72	+2.79	5.00	5.64	10.66	48.59
3	9,472	96.56	+4.95	8.14	9.79	17.64	41.09
4	9,472	98.52	+7.07	12.32	14.02	24.97	29.08
5	9,472	99.33	+9.18	18.12	19.31	33.97	22.69
All	47,360	95.15	+5.07	9.27	10.18	18.47	39.29

G.3. Additive composition across retrieval cutoffs

The central composition question is whether δ_3 can be reconstructed from $\delta_1 + \delta_2$. Table S14 reports signed additive error and condition-level R^2 across Recall@1, Recall@5, and Recall@10.

At mild severity the two marginals reconstruct the joint

condition closely at every cutoff. At severity 5, however, the direction of additive bias differs by metric: Recall@1 is subadditive (−3.46 points), Recall@5 is nearly additive (+0.11), and Recall@10 is mildly above the sum (+1.34). The high-severity Recall@1 pattern is therefore consistent with a top-rank floor effect, not a universal property of joint

Table S13. Primary Recall@1 result by dataset and retrieval direction.

Stratum	Exceed %	Mean excess
MS-COCO I2T	96.49	+5.48
MS-COCO T2I	96.04	+3.12
Flickr30K I2T	91.27	+6.44
Flickr30K T2I	96.69	+5.31

corruption. Across the full primary subset, mean additive error is -0.99 points at Recall@1, $+0.30$ at Recall@5, and $+0.70$ at Recall@10.

G.4. Realized-damage matching

Common ordinal severity does not imply matched realized strength. We therefore restrict conditions by $|\delta_1 - \delta_2|$, measured directly in Recall@1 points, using progressively wider matching tolerances.

The principal result therefore persists even when the two marginals are similar in realized retrieval damage. Equal ordinal severity is a convenient benchmark protocol, not the source of the observed joint-corruption effect.

G.5. Model-level heterogeneity and ranking

The full 19-model condition table appears in the main paper. Here we emphasize the additional cross-model summaries used to interpret it. At Recall@1 severity 5, additive prediction error ranges from -7.04 points (OpenCLIP ViT-B/32) to -0.58 (DFN ViT-H/14). The across-model standard deviation of additive error increases from 0.19 at severity 1 to 0.37, 0.68, 1.25, and 2.03 points through severity 5. Within every family containing multiple variants, the higher-capacity or higher-compute configurations show less negative severity-5 error; these comparisons are descriptive rather than controlled scaling experiments.

Image-only degradation tracks the joint model ranking strongly (Spearman $\rho = 0.912$, 95% model-bootstrap CI $[0.715, 0.984]$), text-only degradation is less predictive ($\rho = 0.512$), and the two marginal rankings agree only weakly ($\rho = 0.221$, $p = 0.363$). Joint degradation as a fraction of clean performance ranges from 36.0% to 20.0% across the roster and correlates negatively with parameter count ($\rho = -0.790$, $p < 0.0001$); this association is not interpreted causally.

H. Statistical Inference and Generalization Sensitivity

H.1. Analysis units

Matched triples are repeated measurements rather than independent draws. Each model appears across many operator pairs, and severity, dataset, and direction recur within a model-operator structure. Condition-level confidence intervals therefore cluster by model $\times$ image corruption $\times$

text perturbation, preserving severity, dataset, and direction within a cluster. The main analyses use 50,000 percentile bootstrap resamples. Model-ranking intervals instead resample the 19 models.

H.2. Fixed-benchmark versus crossed-axis sensitivity

The clustered intervals in the paper quantify uncertainty conditional on the observed benchmark structure; they are not a claim of generalization to arbitrary unseen models or corruption operators. As a stricter sensitivity analysis we separately resample the model, image-corruption, and text-perturbation axes.

The broader resampling widens uncertainty substantially but preserves the qualitative conclusion for both the primary and challenge subsets. We treat this as sensitivity, not as a theorem about arbitrary unseen models or interventions.

I. Recovery Under Joint Corruption

The recovery study is a downstream stress test, not the defining benchmark contribution. Unless otherwise stated, these experiments use MS-COCO and the seven-operator high- damage challenge subset.

I.1. Complete adaptation stress-test suite

The suite contains two lightweight procedures defined for this study—**Tandem-Aug** and **Tandem-Align**—plus QueryExpansion [4], FeatureWhitening [7], and gradient-based TENT [19], MEMO [23], TPT [17], and DiffTPT [5]. Tandem-Aug averages image representations across augmented views without updating model parameters. Tandem-Align is a closed-form feature-space procedure that shifts and scales text features to match the image-feature mean and standard deviation before re-normalization; it performs no prompt optimization and is not attributed to a published prompt-alignment method.

Coverage differs across methods. QueryExpansion, Tandem-Aug, and FeatureWhitening cover all seven general-suite checkpoints; Tandem-Align covers five compatible checkpoints; the gradient methods have one completed checkpoint in the pooled table. These pooled numbers are therefore interpreted within method rather than as a leaderboard.

Implementation configurations. Tandem-Aug uses eight augmented views plus the original (horizontal flip, RandomResizedCrop at scale 0.7–1.0, and ColorJitter 0.2/0.2/0.1/0.05), then averages the nine image features; text is unchanged. TENT updates LayerNorm affine parameters with Adam (lr 10^{-3} , three steps, batch 64) using entropy minimization. MEMO updates all parameters per image using eight views, Adam (lr 10^{-4}), three steps,

Table S14. Additive composition across retrieval cutoffs on the execution-screened primary subset. Bias is $\delta_3 - (\delta_1 + \delta_2)$.

S	Recall@1		Recall@5		Recall@10	
	Bias	R^2	Bias	R^2	Bias	R^2
1	+0.15	.968	+0.10	.980	+0.12	.980
2	+0.03	.970	+0.25	.980	+0.33	.982
3	-0.28	.948	+0.52	.969	+0.67	.973
4	-1.38	.887	+0.52	.950	+1.06	.956
5	-3.46	.759	+0.11	.895	+1.34	.918

Table S15. Post-hoc realized-damage matching on the execution-screened primary subset. This is a retrieval-damage sensitivity analysis, not a physical or perceptual calibration of corruption strength.

Matching rule	n	Exceed %	Mean excess	Mean additive error
$ \delta_1 - \delta_2 \leq 0.5$ pp	3,229	96.97	+4.61	-0.18
≤ 1 pp	6,302	96.45	+4.55	-0.21
≤ 2 pp	11,855	95.39	+4.44	-0.26
≤ 3 pp	16,370	94.47	+4.51	-0.34
≤ 5 pp	23,261	94.23	+4.79	-0.48
≤ 10 pp	35,014	94.77	+5.04	-0.69

Table S16. Bootstrap sensitivity. “Clustered” conditions on the observed benchmark structure; “crossed-axis” separately resamples models, image operators, and text operators.

Subset	Statistic	Point	Clustered 95% CI	Crossed-axis 95% CI
Execution primary	Mean excess	+5.07	[4.96, 5.17]	[3.64, 6.41]
Execution primary	Exceedance %	95.15	[94.78, 95.52]	[91.46, 98.15]
Challenge	Mean excess	+5.63	[5.52, 5.72]	[4.50, 6.66]
Challenge	Exceedance %	96.15	[95.77, 96.50]	[92.69, 98.63]

Table S17. Complete adaptation stress-test suite on COCO joint corruption using the seven-operator high-damage challenge subset. Coverage differs across methods, so pooled magnitudes are interpreted within method rather than as a cross-method leaderboard.

Method	What it adapts	I2T	95% CI	T2I	95% CI	Coverage
TENT	LayerNorm affines	+3.51	[+3.14, +3.89]	+1.67	[+1.54, +1.79]	1/7
QueryExpansion	none (query side)	+3.40	[+3.24, +3.55]	-8.72	[-8.85, -8.60]	7/7
MEMO	full model, per-image	+2.33	[+2.20, +2.46]	-21.29	[-21.80, -20.78]	1/7
Tandem-Aug	none (augment + average)	+0.83	[+0.74, +0.91]	+1.65	[+1.60, +1.70]	7/7
TPT	prompts	-3.27	[-3.63, -2.93]	-2.91	[-3.05, -2.76]	1/7
DiffTPT	prompts (strong aug)	-3.76	[-4.14, -3.39]	-2.79	[-2.92, -2.67]	1/7
Tandem-Align	feature-statistic alignment	-13.28	[-13.67, -12.88]	-12.70	[-12.93, -12.46]	5/7
FeatureWhitening	embedding space	-25.75	[-26.24, -25.27]	-24.21	[-24.54, -23.87]	7/7

327 and marginal entropy. TPT updates prompt embeddings
 328 using eight views, Adam ($\text{lr } 5 \times 10^{-3}$), three steps, and
 329 the lowest-entropy 10% of views; DiffTPT uses the same
 330 optimizer with stronger crops, color jitter, grayscale, and
 331 Gaussian blur. Tandem-Align performs only closed-form
 332 text-feature mean/standard-deviation matching followed
 333 by re-normalization. FeatureWhitening applies ZCA
 334 whitening to gallery features (eigenvalue floor 10^{-5}).
 335 QueryExpansion uses the top 10 neighbours and refines
 336 the query as $0.5q + 0.5\bar{n}$. TCR uses the authors’ released
 337 query-tower LayerNorm parameter scope and defaults.
 338 Exact implementations and provenance are included in the
 339 release.

I.2. Controlled EVA-CLIP ViT-L/14 comparison 340

A single-backbone comparison removes the coverage con- 341
 found for the general suite. 342

Only TENT and Tandem-Aug improve both directions 343
 in this controlled comparison. The mechanism behind 344
 the large negative effects of several stronger representa- 345
 tion/prompt updates is outside the scope of this benchmark 346
 and should not be inferred from these results alone. 347

I.3. TCR under query, gallery, and joint corruption 348

TCR [10] was developed for query shift with a clean gallery. 349
 We therefore separate corrupted-query/clean-gallery, clean- 350
 query/corrupted-gallery, and joint regimes. 351

For I2T, joint corruption retains a substantial +4.73-point 352
 gain. The pooled T2I joint effect is near zero, but the 353
 checkpoint-level decomposition is heterogeneous: 354

Table S18. Controlled comparison on EVA-CLIP ViT-L/14 under identical COCO joint-corruption conditions from the high-damage challenge subset. Values are $\Delta\text{Recall}@1$ relative to no adaptation.

Method	I2T	T2I	Both > 0
TENT	+3.51	+1.67	yes
MEMO	+2.33	-21.29	no
QueryExpansion	+1.88	-11.69	no
Tandem-Aug	+1.47	+1.83	yes
TPT	-3.27	-2.91	no
DiffTPT	-3.76	-2.79	no
Tandem-Align	-13.94	-18.33	no
FeatureWhitening	-37.82	-31.08	no

Table S19. TCR by corruption regime. Query shift is the published clean-gallery setting; joint corruption degrades both retrieval sides.

Regime	I2T	95% CI	T2I	95% CI
query shift	+5.90	[+5.07, +6.75]	-0.30	[-0.82, +0.21]
gallery shift	+1.83	[+1.40, +2.21]	+0.46	[+0.11, +0.84]
joint	+4.73	[+4.36, +5.10]	-0.03	[-0.23, +0.17]

All three evaluated OpenAI CLIP checkpoints improve in T2I (+1.11 to +1.79), while all seven other complete checkpoints decline (-0.21 to -1.10). I2T gains remain positive for all ten completed checkpoints. Because architecture, tokenizer, objective, and pretraining corpus covary, this split is descriptive rather than causal.

I.4. Absolute and fractional recovery

Absolute recovery can increase simply because more performance has been lost. For TCR, I2T gain rises from +3.17 points at severity 1 to +4.78 at severity 3 and +6.22 at severity 5, while fractional recovery falls from 48.6% to 18.8% to 11.5%. Tandem-Aug shows the same decline in recovered fraction: 45.5%, 18.5%, and 8.1%. Reporting both prevents increasing point gain from being mistaken for an increasing fraction of corruption damage repaired.

I.5. Input restoration

We complement model-side adaptation with direct input restoration on five models, 12 corruption pairs, and severities 1, 3, and 5.

Pooled across conditions, restoring the image recovers +0.25 Recall@1 points, restoring the caption +1.72, and restoring both +2.00. Their interaction is +0.02 [$-0.03, +0.07$], giving no measurable gain beyond the separate restoration effects. Restoration is corruption-specific: spelling correction recovers +6.36 points for keyboard typos but changes performance by -0.74 on object substitution and -0.44 on word deletion, where it does not invert the underlying transformation.

J. Evaluation Protocol, Reproducibility, and Release

J.1. Expanded joint-corruption evaluation protocol

The main paper ships a seven-point protocol. Here we state the same protocol with the benchmark artifact associated with each step.

- Check applicability and execution.** Report whether the structure required by an operator exists and whether the requested transformation actually changes the input. TandemBench provides applicability/fire-rate tables and inspectable before/after text outputs.
- Separate execution from difficulty.** Treat marginal degradation as benchmark difficulty rather than a validity test. Damage-based subsets should be named explicitly as challenge/stress-test subsets.
- Match marginal and joint conditions.** Measure image-only, text-only, and joint corruption on the same samples and clean baseline. If an ordinal severity index is used, do not equate index equality with equal realized damage.
- Keep $\delta_1, \delta_2, \delta_3$ explicit.** Report raw degradation terms before normalized interaction summaries; use a denominator safeguard when marginal damage is near zero.
- Use both joint references.** Compare δ_3 with $\max(\delta_1, \delta_2)$ for degradation missed by either marginal alone and with $\delta_1 + \delta_2$ for marginal-to-joint reconstruction.
- Resolve composition.** Report behavior across severity and retrieval metric and inspect dataset/direction/model variation rather than relying only on one pooled statistic.
- Test recovery jointly.** When adaptation/restoration is studied, report retrieval direction, model coverage, absolute gain, and recovered fraction under the condition in which both retrieval sides are degraded.

J.2. Release contents

The release supports two reproduction levels. Full re-evaluation uses corruption implementations, model configurations, deterministic generation scripts, and recovery code to rerun the benchmark from public source datasets and referenced checkpoints. Corrupted image collections are not redistributed. Analysis-only reproduction uses the complete evaluation records, screening records, statistical scripts, table generators, and metadata without rerunning neural models.

The release contains all 83,754 evaluation records and deterministic configurations for reconstructing the corrupted image conditions. Keeping corrupted images out of the archive avoids redistributing derived copies of the source datasets while preserving the evaluation procedure.

Table S20. TCR under joint corruption by checkpoint. The pooled T2I null combines two consistent groups: the three OpenAI CLIP checkpoints improve, while all seven other complete checkpoints decline.

Model	Pretraining	I2T	95% CI	T2I	95% CI
clip_vitb16	OpenAI WIT	+4.41	[+3.47, +5.46]	+1.79	[+1.66, +1.91]
clip_vitb32	OpenAI WIT	+3.46	[+2.40, +4.60]	+1.11	[+0.94, +1.28]
clip_vitl14	OpenAI WIT	+4.30	[+3.37, +5.14]	+1.57	[+1.45, +1.70]
evaclip_vitb16	Merged-2B	+3.22	[+2.29, +4.16]	-0.69	[-0.82, -0.57]
metaclip_vitb16	MetaCLIP	+5.37	[+4.05, +6.85]	-0.47	[-0.61, -0.35]
metaclip_vitl14	MetaCLIP	+5.81	[+4.69, +6.85]	-1.01	[-1.32, -0.71]
openclip_vitb16	LAION-400M	+5.38	[+4.39, +6.33]	-0.28	[-0.46, -0.11]
openclip_vitb32	LAION-2B	+5.22	[+4.18, +6.25]	-0.21	[-0.36, -0.07]
openclip_vitl14	LAION-2B	+5.26	[+4.15, +6.44]	-1.02	[-1.43, -0.67]
siglip_vitb16	WebLI	+4.81	[+3.79, +5.72]	-1.10	[-1.43, -0.83]

Table S21. Input-restoration results. Values are Recall@1 recovery relative to unrestored corrupted inputs. Interaction is the gain from restoring both beyond the sum of the two separate restoration gains.

Group	Img	95% CI	Text	95% CI	Both	95% CI	Inter.	95% CI
all	+0.25	[+0.09, +0.41]	+1.72	[+0.88, +2.62]	+2.00	[+1.12, +2.93]	+0.02	[-0.03, +0.07]
severity 1	-0.38	[-0.55, -0.21]	-0.08	[-0.32, +0.18]	-0.43	[-0.72, -0.13]	+0.02	[-0.04, +0.08]
severity 3	+0.35	[+0.18, +0.52]	+2.05	[+1.09, +3.05]	+2.37	[+1.38, +3.40]	-0.03	[-0.11, +0.04]
severity 5	+0.78	[+0.53, +1.05]	+3.20	[+1.83, +4.65]	+4.07	[+2.62, +5.59]	+0.08	[-0.01, +0.18]
keyboard.typo	+0.32	[+0.03, +0.62]	+6.36	[+5.63, +7.15]	+6.74	[+5.88, +7.64]	+0.06	[-0.04, +0.18]
object.substitution	+0.21	[-0.05, +0.47]	-0.74	[-0.85, -0.64]	-0.50	[-0.77, -0.24]	+0.03	[-0.04, +0.10]
word.delete	+0.23	[-0.05, +0.51]	-0.44	[-0.54, -0.34]	-0.24	[-0.50, +0.03]	-0.03	[-0.09, +0.05]

J.3. Implementation provenance

Tandem-Aug and Tandem-Align are study procedures introduced only as recovery probes. Published methods retain their citations and implementation provenance. In particular, TCR uses the authors’ released parameter scope/defaults and QueryExpansion follows the specified nearest-neighbour refinement. Tandem-Align is an internal study name for this closed-form feature-statistic procedure and carries no external method attribution.

J.4. Computational environments

Experiments were distributed according to method requirements. The main 19-model benchmark, corruption generation, screening/analysis, non-gradient recovery probes, table generation, and audits ran on an Apple M4 Pro system (24 GB unified memory, macOS 26.5.2; Python 3.13.5, PyTorch 2.8.0/MPS, open_clip 3.3.0, transformers 5.14.1, NumPy 2.1.3, SciPy 1.15.3). Gradient-based TENT, MEMO, TPT, and DiffTPT runs used a Linux Ubuntu 24.04.3 server with a Threadripper PRO 7965WX, 125 GiB RAM, and three RTX PRO 6000 Blackwell Max-Q GPUs. TCR and input restoration used a Windows 11 workstation with a Core Ultra 7 265K, 63 GB RAM, and an RTX 5000 Ada. Unrecorded Linux/Windows package versions and an aggregate end-to-end runtime are intentionally not reconstructed from memory.

K. Scope and Limitations

Evaluation scope. TandemBench traces the diagonal of a five-level image–text ordinal severity grid, not the full

asymmetric 5×5 surface. The realized-damage analysis tests sensitivity to the ordinal pairing but is not a physical or perceptual calibration. COCO and Flickr30K have relatively short English captions, and all 19 evaluated systems are dual encoders. Longer documents, multilingual retrieval, cross-encoders, reranking systems, generative VLMs, and temporally structured/video retrieval remain outside the present scope.

Perturbation scope. The corruption suite provides deterministic controlled interventions rather than an exhaustive model of natural deployment shift. The execution screen measures whether an operator actually acts on these captions; it is not a certification of semantic naturalness, meaning preservation, or perceptual realism. A human semantic/perceptual validity audit was not performed. Structure-dependent semantic operators remain in the full suite even when they are rarely applicable; an eligibility-conditioned rerun on only structurally eligible captions would require new inference and is left for future work.

Inference scope. Primary clustered intervals describe uncertainty conditional on the fixed benchmark structure. Crossed-axis resampling is a broader sensitivity analysis, but neither result justifies a claim about arbitrary unseen models or corruption operators. Model-size and family trends are descriptive because architecture, objective, compute, tokenization, and sometimes pretraining data differ across checkpoints.

Recovery scope. The recovery experiment is a stress test with unequal method coverage, not a comprehensive TTA leaderboard. The controlled EVA-CLIP comparison removes the coverage confound for one backbone only. The observed directional/model-family differences motivate dedicated adaptation work but do not identify a causal mechanism.

Release boundary. Corrupted image sets are not hosted. Deterministic generation scripts and configurations reconstruct them from public source data, while the released evaluation records preserve the complete benchmark outputs used for analysis.

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